

EXECUTIVE SUMMARY

Carbon Sequestration & Carbon Credit Analysis

Golden Forests / CRASSNA Agroforestry - 25 Year Projection

KEY PROJECT METRICS

Total CO ₂ Sequestered (25 Years)	1,160,077 tonnes
Total Carbon Credit Revenue	\$28,355,642 USD
Net Revenue After Certification	\$27.6M - \$27.9M USD
Project Duration	25 years 2026-2050
Location	Philippines
Land Type	Degraded agricultural land

CO₂ SEQUESTRATION BY SPECIES (Maturity-Adjusted)

Agarwood (<i>Aquilaria crassna</i>)	208,466 tonnes	18.0%
Mango (Dwarf variety)	383,622 tonnes	33.1%
Native Trees (4-species mix)	567,990 tonnes	49.0%
TOTAL	1,160,077 tonnes	100%

CARBON CREDIT REVENUE BY SPECIES (Maturity-Adjusted)

Agarwood	\$4,752,714	16.8%
Mango	\$9,459,996	33.4%
Native Trees	\$14,142,932	49.9%
TOTAL	\$28,355,642	100%

CARBON SEQUESTRATION RATES BY SPECIES

Agarwood	<i>Aquilaria crassna</i>	23.5
Mango (Dwarf)	<i>Mangifera indica</i>	50
Native Mix (average)	(4-species)	23
— Narra	<i>Pterocarpus indicus</i>	22.5
— Agoho	<i>Casuarina equisetifolia</i>	32.5
— Molave	<i>Vitex parviflora</i>	17.5
— Talisay	<i>Terminalia catappa</i>	20

CARBON CREDIT PRICING

Base Price (UAE Voluntary Market)	\$15 per tonne CO ₂
Annual Price Escalation	3% per annum
Market Comparison (Sept 2025)	\$24 ARR credits trading
Forward Pricing (2026+)	\$50+ quality forestry credits

Note: \$15/tonne base assumption is conservative (~37% below current market rates)

CERTIFICATION & VERIFICATION COSTS

Initial Certification (Verra VCS / Gold Standard)	\$50,000 - \$100,000 One-time
Project Design Document (PDD)	\$20,000 - \$40,000 One-time
Annual Monitoring & Verification	\$15,000 - \$25,000 Per year
Registry Fees	\$0.10 - \$0.20 Per credit issued
Total 25-Year Estimate	\$450,000 - \$750,000 All costs
NET REVENUE AFTER CERTIFICATION	\$27.6M - \$27.9M Highly attractive

METHODOLOGY & COMPLIANCE

- Follows IPCC LULUCF Guidelines for carbon accounting
- Certification Compatible: Verra VCS, Gold Standard, Plan Vivo
- Age-adjusted (maturity-weighted) sequestration rates applied
- Conservative assumptions throughout for defensible projections

PROJECT GROWTH TRAJECTORY

Milestone	Standing Trees	Annual Revenue
Year 1 (2026)	75,502	\$2,312
Year 5 (2030)	642,212	\$95,848
Year 10 (2035)	1,689,483	\$676,050
Year 15 (2040) - Peak Trees	3,124,326	\$1,922,398
Year 25 (2050)	2,174,287	\$1,934,960

KEY INVESTMENT HIGHLIGHTS

Peak Annual Revenue	\$1,947,038	2041
Peak Standing Trees	3,124,326	2040
Revenue Growth (Year 1→10)	292x	multiple
Average Annual Revenue (Years 10-25)	\$1,681,934	

REVENUE TIMELINE

- Years 1-2: Project registration & baseline establishment — No credits issued
- Years 3-4: First monitoring period — Initial credit issuance begins
- Years 5+: Annual credit issuance based on verified sequestration
- Forward Sales Option: Pre-sell future vintage credits for earlier cash flow

RISK FACTORS & MITIGATION

Risk Factor	Mitigation
Permanence Risk	Buffer pool (10-20% credits reserved); conservative rates provide implicit buffer
Additionality	Project scale requires carbon revenue; baseline is degraded land use
Leakage	Uses degraded/agricultural land; no displacement of forest or communities

SOURCES & REFERENCES

Carbon Sequestration Rates:

- Agarwood (23.5 kg CO₂/tree/yr)
- Mango (50 kg CO₂/tree/yr)
- Native Trees (Narra, Agoho, Molave, Talisay)

<i>Oud Academia (2025)</i>	oudacademia.com
<i>Trivedi et al. (2025), IEAM Vol.21(5)</i>	doi.org/10.1093/inteam/vjaf045
<i>Philippine Forestry Studies</i>	Pansit 2019, Madalmingan 2024

Methodology & Standards:

- Carbon Accounting
- Certification Standards

<i>IPCC LULUCF Guidelines</i>	ipcc-nggip.iges.or.jp
<i>Verra VCS, Gold Standard, Plan Vivo</i>	verra.org , goldstandard.org

Carbon Credit Pricing:

<i>Sylvera VCM Q3 2025 Report</i>	UAE Voluntary Carbon Market
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